

Introduction

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Advanced Macroeconomics I

Two Workhorse Classes of Models

1. Standard Incomplete Markets Model

- Bewley (1977), Imrohoroglu (1987), Huggett (1993), Aiyagari (1994)

2. Firm Dynamics

- Lucas (1978), Jovanovic (1982), Hopenhayn (1992)
- Most research builds on these by adding various frictions
- Today: review models, then numerical methods

Households

- Preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{c_{it}^{1-\theta}}{1-\theta} - \frac{h_{it}^{1+\gamma}}{1+\gamma} \right)$$

- Budget constraint

$$c_{it} + b_{it+1} + k_{it+1} + Q_t s_{it+1} = T_t + (1 - \tau) W_t e_{it} h_{it} + (1 + r_{t-1}) b_{it} + (1 - \delta + R_t) k_{it} + (Q_t + D_t) s_{it}$$

- Idiosyncratic ability follows AR(1), no aggregate uncertainty

$$\log e_{it} = \rho \log e_{it-1} + \sigma_e \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, 1)$$

- Lump-sum transfer T_t financed by labor income τ . Save in
 - government bonds b_{it+1} with interest rate r_t
 - physical capital k_{it+1} with depreciation δ and rental rate R_t
 - firm shares s_{it+1} with price Q_t and dividends D_t

Household Choices

- Since no aggregate uncertainty, no-arbitrage implies

$$1 + r_t = 1 - \delta + R_{t+1} = \frac{Q_{t+1} + D_{t+1}}{Q_t}$$

- so portfolio choice indeterminate
 - with aggregate uncertainty, interesting portfolio choice problem and puzzles
- Budget constraint ($a_{t+1} = b_{it+1} + k_{it+1} + Q_t s_{it+1}$)

$$c_{it} + a_{it+1} = T_t + (1 - \tau) W_t e_{it} h_t + (1 + r_{t-1}) a_{it}$$

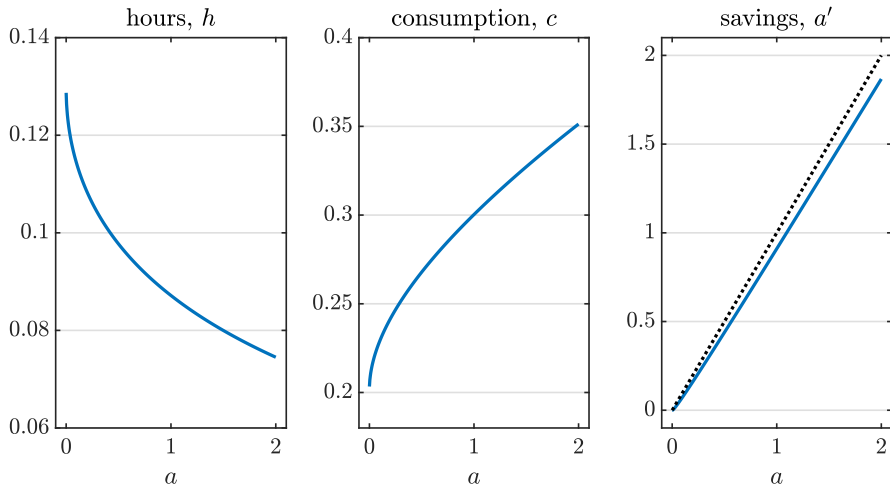
- Assume borrowing limit, $a_{t+1} \geq \bar{a}$ so Euler equation

$$c_{it}^{-\theta} = \beta \mathbb{E}_t c_{it+1}^{-\theta} (1 + r_t + \xi_{it})$$

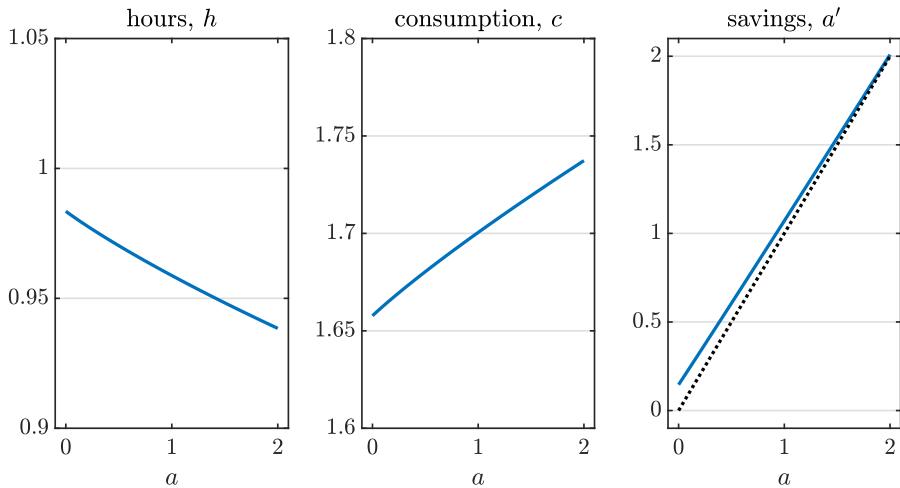
- Hours Foc:

$$h_{it}^{\gamma} = (1 - \tau) W_t e_{it} c_{it}^{-\theta}$$

Decision Rules: Low Ability



Decision Rules: High Ability



Aggregate Labor Supply

- Useful to aggregate decision rules
- Let $\omega_{it} = c_{it}/C_t$ – consumption share of i (time-invariant in complete markets)
- Labor Foc

$$h_{it} = ((1 - \tau) W_t)^{\frac{1}{\gamma}} C_t^{-\frac{\theta}{\gamma}} e_{it}^{\frac{1}{\gamma}} \omega_{it}^{-\frac{\theta}{\gamma}}$$

- Aggregate across all agents ($L_t = \int e_{it} h_{it} di$)

$$L_t^\gamma = \lambda_t W_t C_t^{-\theta}, \quad \lambda_t = (1 - \tau) \left(\int \omega_{it}^{-\frac{\theta}{\gamma}} e_{it}^{1+\frac{1}{\gamma}} di \right)^\gamma$$

- incomplete markets $\rightarrow$ time-varying wedge in aggregate labor supply
- e.g., tightening of credit limit increases ω_i for high e_i households, reducing λ_t

Aggregate Euler Equation

- Individual Euler equation

$$1 = \mathbb{E}_t \left(\frac{\omega_{it+1} C_{t+1}}{\omega_{it} C_t} \right)^{-\theta} (1 + r_t + \xi_{it})$$

- Aggregate across all agents ($\zeta_{it} = \frac{1+r_t+\xi_{it}}{1+r_t}$)

$$1 = \beta \zeta_t \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} (1 + r_t), \quad \zeta_t = \int \mathbb{E}_t \left(\frac{\omega_{it+1}}{\omega_{it}} \right)^{-\theta} \zeta_{it} di$$

- since $u'(c) = c^{-\theta}$ convex ($u'''(c) > 0$), $\zeta_t > 1$
- tightening of credit, or stronger precautionary savings: increase ζ_t

Firms

- Mass N_t , pinned down by a free entry condition
- Heterogeneous in productivity z_{it} , exogenous exit at rate φ
- Two approaches to ensure non-degenerate size distribution
 1. decreasing returns to scale (today)
 2. differentiated goods (later in course)

Firm Problem

- Each firm has production function

$$y_{it} = z_{it} (k_{it}^\alpha l_{it}^{1-\alpha})^\eta$$

- Ex-dividend value

$$Q_{it} = \frac{1 - \varphi}{1 + r_t} \mathbb{E}_t (Q_{it+1} + D_{it+1}), \quad D_{it} = z_{it} (k_{it}^\alpha l_{it}^{1-\alpha})^\eta - W_t l_{it} - R_t k_{it}$$

- Focs for capital and labor:

$$\alpha \eta \frac{y_{it}}{k_{it}} = R_t, \quad (1 - \alpha) \eta \frac{y_{it}}{l_{it}} = W_t$$

- Implies

$$y_{it} = z_{it}^{\frac{1}{1-\eta}} \left[\frac{1}{\eta} \left(\frac{R_t}{\alpha} \right)^\alpha \left(\frac{W_t}{1-\alpha} \right)^{1-\alpha} \right]^{-\frac{\eta}{1-\eta}}$$

Aggregation

- Let $K_t = \int_0^{N_t} k_{it} di$ etc.

- Gives aggregate production function

$$Y_t = Z_t (K_t^\alpha L_t^{1-\alpha})^\eta$$

- Demand for capital and labor:

$$R_t = \alpha \eta \frac{Y_t}{K_t} \quad \text{and} \quad W_t = (1 - \alpha) \eta \frac{Y_t}{L_t}$$

- Aggregate productivity:

$$Z_t = \left(\int_0^{N_t} z_{it}^{\frac{1}{1-\eta}} \right)^{1-\eta} di = N_t^{1-\eta} \left(\int z^{\frac{1}{1-\eta}} d\mathcal{F}(z) \right)^{1-\eta}$$

- Increases with N_t because of decreasing returns at the firm level

Free entry

- Let $Q_t = \int Q_t(z) d\mathcal{F}(z)$ claim to diversified portfolio of firm of mass 1
- Assume cost κ to create new firm, draw z from invariant distribution $\mathcal{F}$
- If positive entry

$$\kappa = Q_t = \frac{1 - \varphi}{1 + r_t} \left[Q_{t+1} + (1 - \eta) \frac{Y_{t+1}}{N_{t+1}} \right]$$

- Q_t is price of a claim to portfolio of all firms, so $\mathbb{E}_t D_{it+1} = (1 - \eta) \frac{Y_{t+1}}{N_{t+1}}$
- Free entry condition pins down mass of new entrants ν_t

$$- N_{t+1} = (1 - \varphi)(N_t + \nu_t)$$

Government

- Labor tax τ finances lump-sum transfer T_t , G and interest on debt
- Budget constraint

$$B_{t+1} + \tau W_t \int e_{it} h_{it} di - T_t = (1 + r_t) B_t + G$$

- Government debt evolves according to

$$\log B_{t+1} = \rho_b \log B_t + (1 - \rho_b) \log \bar{B} + \varepsilon_{b,t}$$

Equilibrium

- Equilibrium prices $R_t = \alpha \frac{Y_t}{K_t}$ and $W_t = (1 - \alpha) \frac{Y_t}{L_t}$
- Agent decision rules $h_t(a, e)$, $a_{t+1}(a, e)$, $c_t(a, e)$
- Distribution $F_t(a, e)$, mass of firms N_t
- Asset market clearing

$$K_{t+1} + B_{t+1} + Q_t(N_t + \nu_t) = \int a_{t+1}(a, e) dF_t(a, e)$$

- Distribution evolves according to

$$F_{t+1}(\mathcal{A}, \mathcal{E}) = \int \mathbb{I}_{a_{t+1}(a, e) \in \mathcal{A}} dQ(e' \in \mathcal{E} | e) dF_t(a, e)$$

Illustrative Example

- Assume CRS ($\eta = 1$), no firm exit
- Period 1 year

θ	relative risk aversion	1
γ	inverse Frisch elasticity	1
α	capital elasticity	1/3
δ	capital depreciation	0.10
$\bar{B}$	gov't debt to GDP	1

Calibration

- Choose discount factor and ability process (ρ_e, σ_e) to match
 - average wealth to income
 - Gini wealth and income inequality

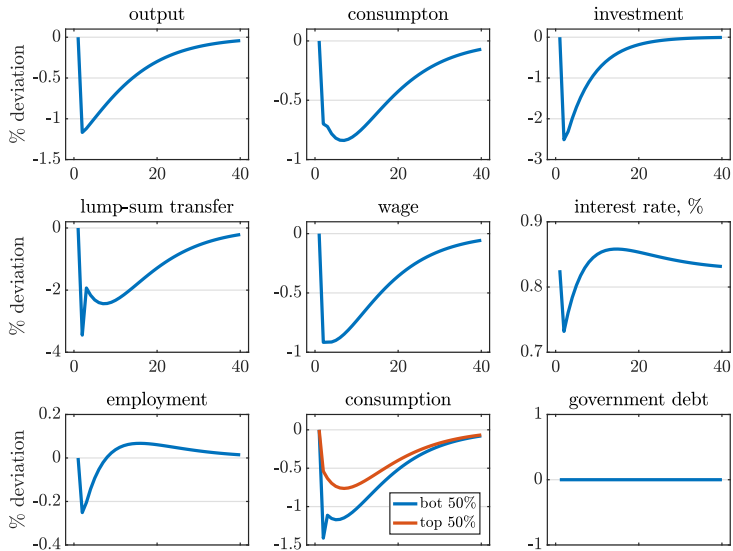
			Data	Model
β	0.978	wealth to income	6.6	6.6
ρ_e	0.987	Gini wealth	0.85	0.86
σ_e	0.175	Gini income	0.64	0.63

- Implies equilibrium interest rate $r = 0.8\%$

Wealth and Income Distribution

	Data	Model		Data	Model
<i>Wealth Distribution</i>			<i>Income Distribution</i>		
Share top 0.1%	0.22	0.04	Share top 0.1%	0.14	0.02
Share top 1%	0.35	0.22	Share top 1%	0.22	0.11
Share top 5%	0.63	0.55	Share top 5%	0.39	0.30
Share top 10%	0.75	0.75	Share top 10%	0.51	0.43
Share bot 75%	0.09	0.05	Share bot 75%	0.29	0.28
Share bot 50%	0.01	0.00	Share bot 50%	0.10	0.10
Share bot 25%	-0.01	0.00	Share bot 25%	0.02	0.01

Response to 1% Productivity Drop



Comparison with Complete Markets Economy

- Same technology parameters, preferences, no heterogeneity

1. $C_t + K_{t+1} - (1 - \delta)K_t = Y_t$

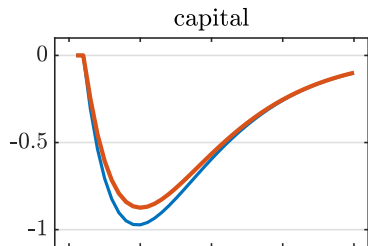
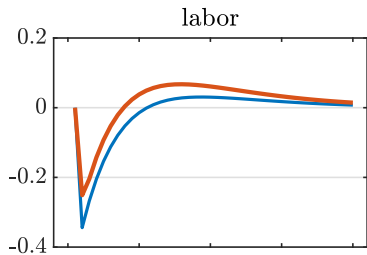
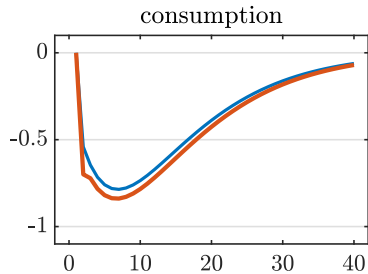
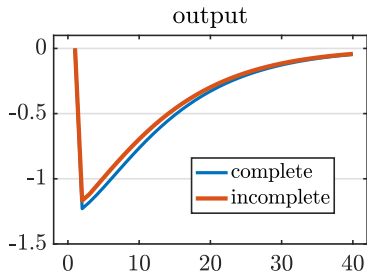
2. $Y_t = Z_t K_t^\alpha L_t^{1-\alpha}$

3. $1 = \beta \mathbb{E}_t \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} \left(\alpha \frac{Y_{t+1}}{K_{t+1}} + 1 - \delta \right)$

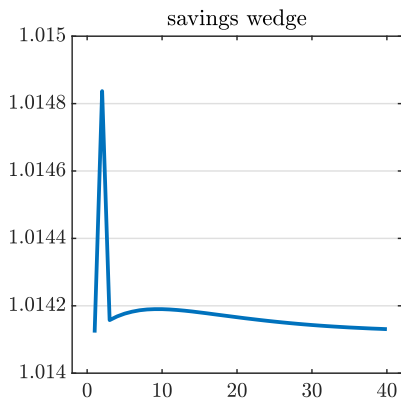
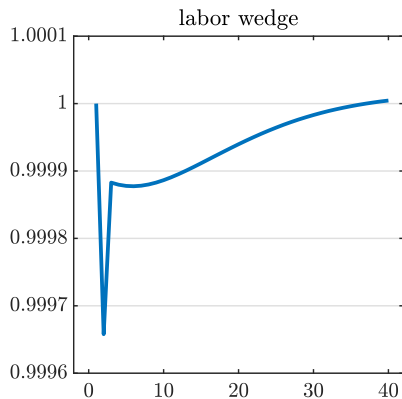
4. $L_t^\gamma = (1 - \alpha) \frac{Y_t}{L_t} C_t^{-\theta}$

- Very different steady-state outcomes, but does the response to shocks differ?

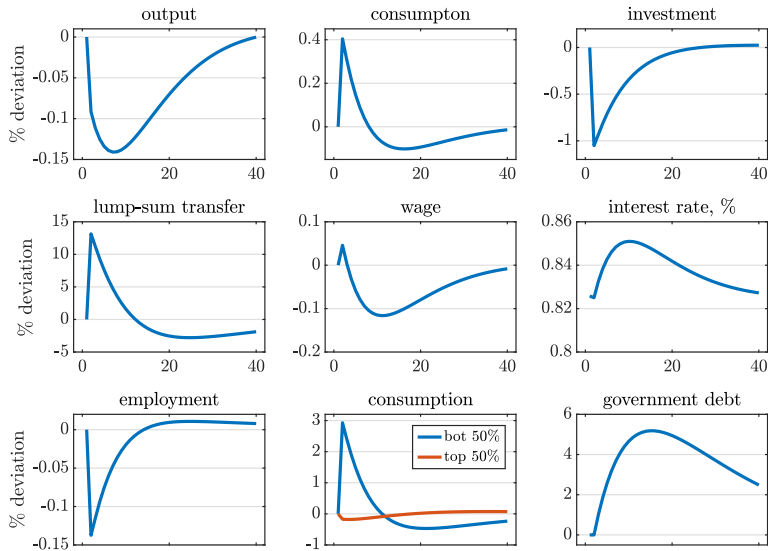
Comparison with Complete Markets Economy



Labor and Savings Wedges



Response to Increase in Gov't Debt



Course Outline

- Computational Tools
 - basics: root-finding, optimization, differentiation, integration
 - functional approximation, solving functional equations
 - projection methods
 - machine learning methods
 - responses to one-time aggregate shocks (sequence space jacobian)
 - Krusell Smith algorithm
- Macro Implications of Household Credit Frictions
 - Guerrieri, Lorenzoni (2017)

Course Outline

- Inequality and Redistribution
 - optimal tax policy in dynamic economies: Boar and Midrigan (2023)
- Firm Dynamics and Misallocation
 - firing frictions: Hopenhayn and Rogerson (1993)
 - size-dependent subsidies: Guner, Ventura, Xu (2008)
 - measurement: Hsieh and Klenow (2009)

Course Outline

- Firm-Level Credit Frictions and Misallocation
 - credit constraints: Buera, Kaboski, Shin (2009), Midrigan-Xu (2013), Moll (2014)
 - uninsurable risk: Boar, Gorea, Midrigan (2023)
- Markups
 - representative consumer: Edmond, Midrigan and Xu (2023)
 - heterogeneity in firm ownership: Boar and Midrigan (2024)

Course Outline

- Menu Cost Models
 - Golosov, Lucas (2007)
 - Midrigan (2010)
 - Alvarez, Lippi and Le Bihan (2014)
 - Boar, Blanco, Jones and Midrigan (2023, 2024)
- Non-Convex Investment Adjustment Costs
 - Khan and Thomas (2008)